

WINNER ENGINEERING LEARNING HUB SERIES

INTERMEDIATE TOPIC

DUCT, PIPE &

EQUIPMENT

SUPPORTING DETAILS



Winner Engineering Pte Ltd
永胜工程私人有限公司
Excellence in Service

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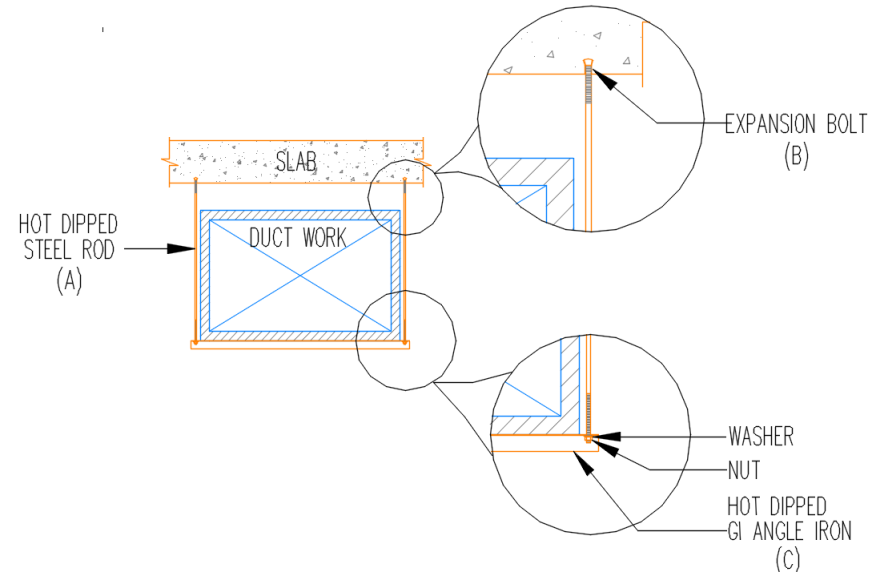
- ▶ Duct Supporting Detail
 - ❖ GI Duct
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 - ❖ CHWP
- ▶ Equipment Supporting Detail
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 - ❖ Cooling Tower
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Duct Supporting Detail:

GI DUCT:

1. Rectangular Duct Support
2. Round Duct Support

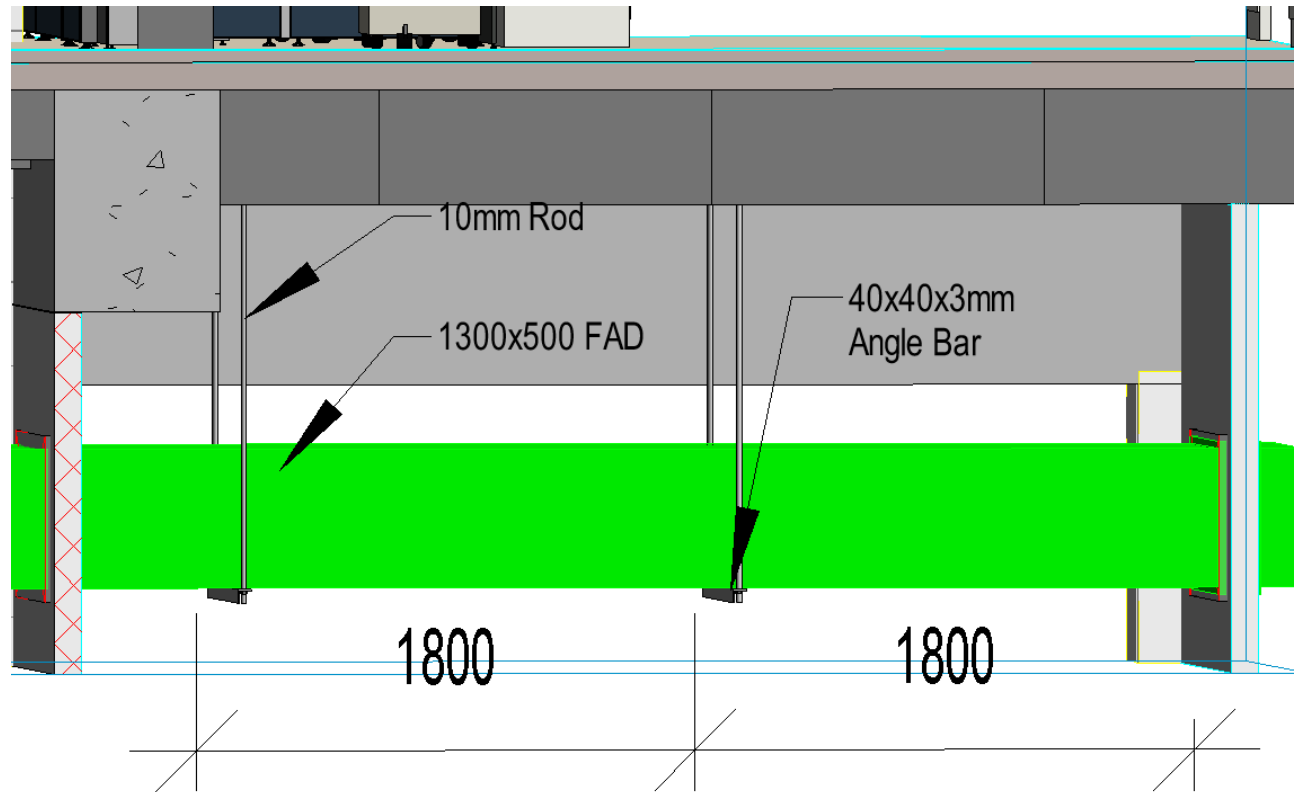
1.Rectangular Duct Support:



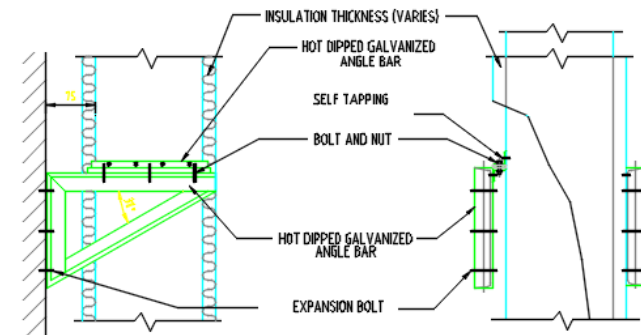
SCHEDULE FOR SUPPORT ROD & ANGLES

DIMENSION OF LONGEST SIDE OF DUCT(mm)	ROD SIZE (A)	EXPANSION BOLT (B)	GI SUPPORT (C)	INTERVAL OF SUPPORT
UP TO 300	DIA. 6mm	DIA. 6mm EXPANSION BOLT	20x20x3 ANGLE BAR	2.4m
325 TO 750	DIA. 6mm	DIA. 6mm EXPANSION BOLT	25x25x3 ANGLE BAR	1.8m
775 TO 1500	DIA. 10mm	DIA. 10mm EXPANSION BOLT	40x40x3 ANGLE BAR	1.8m
1525 TO 2100	DIA. 10mm	DIA. 10mm EXPANSION BOLT	50x50x5 ANGLE BAR	1.8m
2125 OVER	DIA. 10mm	DIA. 10mm EXPANSION BOLT	50x50x6 ANGLE BAR	1.5m

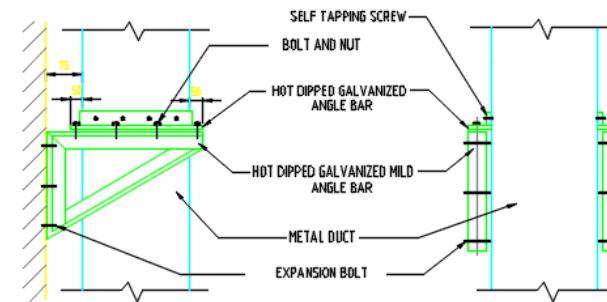
1.Rectangular Duct:



Riser Duct:

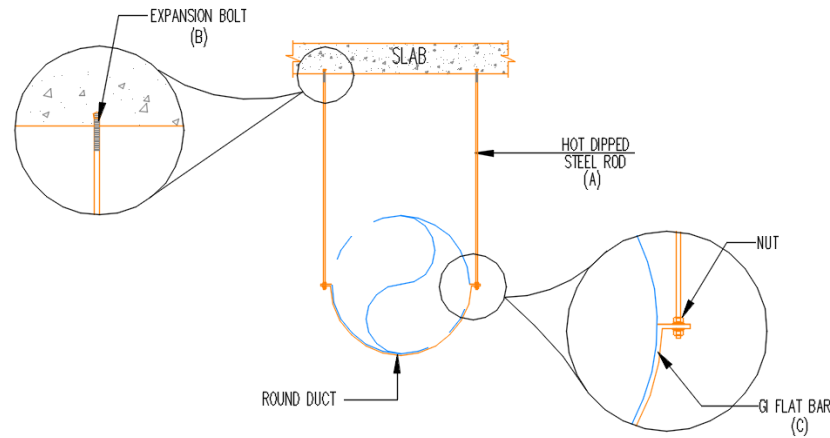


WALL BRACKET FOR INSULATED DUCT



WALL BRACKET FOR NON-INSULATED DUCT

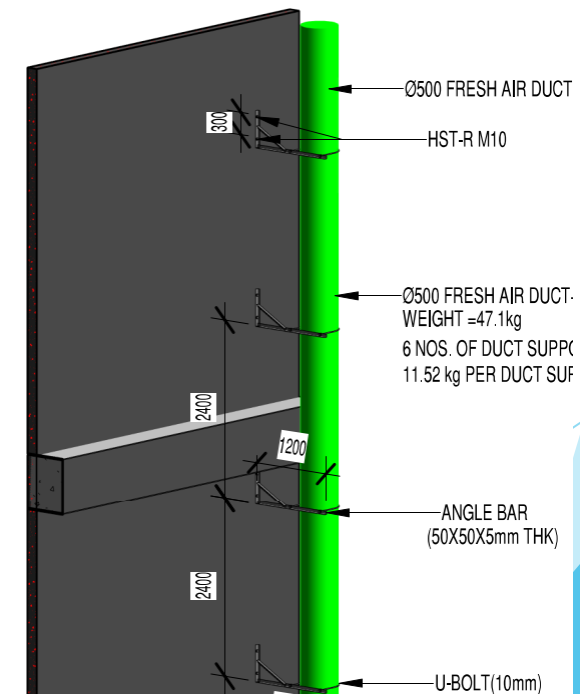
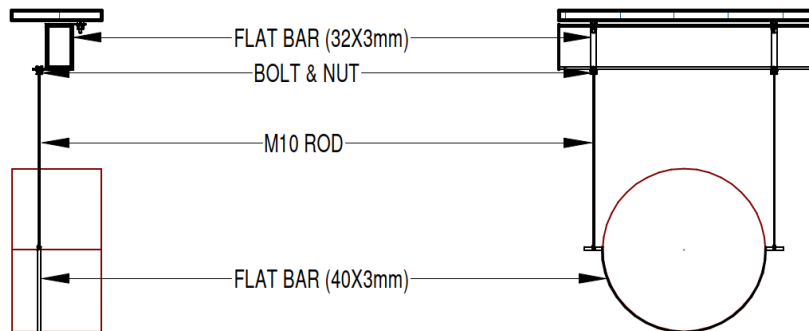
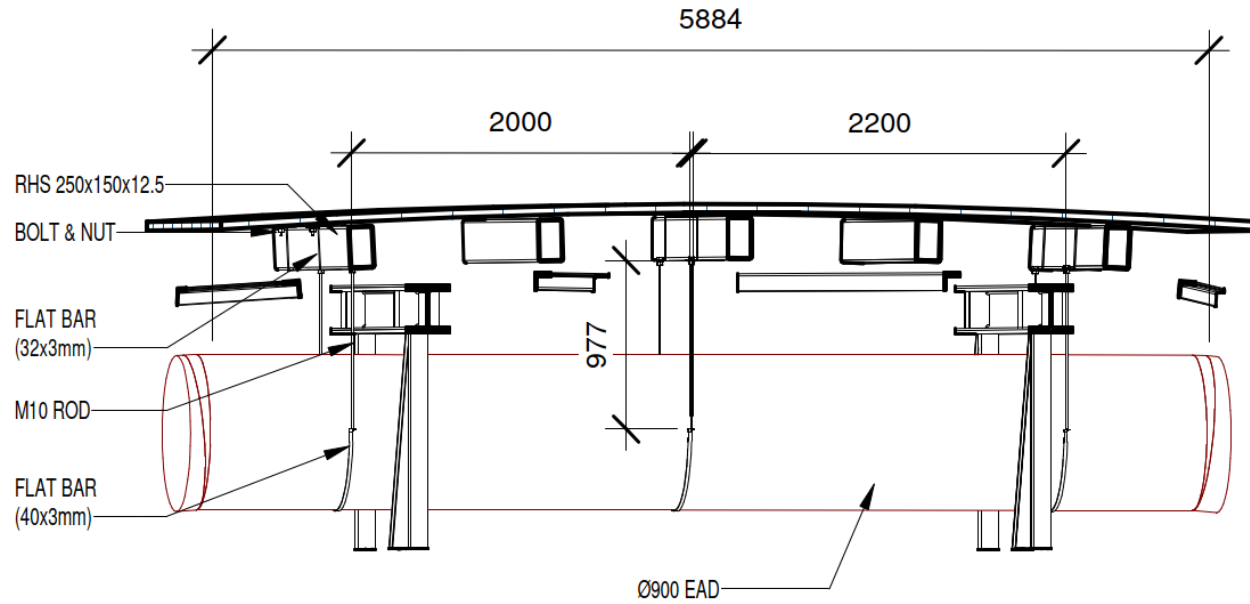
2.Round Duct



SCHEDULE FOR SUPPORT ROD & ANGLES

DIAMETER OF DUCT(mm)	ROD SIZE (A)	EXPANSION BOLT (B)	FLAT BAR (C)	INTERVAL OF SUPPORT
UP TO 250	DIA. 6mm	DIA. 6mm EXPANSION BOLT	40X3 mm	2.4m
450	DIA. 6mm	DIA. 6mm EXPANSION BOLT	40X3 mm	2.4m
600	DIA. 6mm	DIA. 6mm EXPANSION BOLT	40X3 mm	2.4m
900	DIA. 10mm	DIA. 10mm EXPANSION BOLT	40X3 mm	2.4m

Round Duct:



PID DUCT:

SCHEDULE FOR SUPPORT ROD & ANGLES

DIMENSION OF LONGEST SIDE OF PID DUCT(mm)	ROD SIZE (A)	EXPANSION BOLT (B)	GI SUPPORT (C)	INTERVAL OF SUPPORT
UP TO 300	DIA. 6mm	DIA. 6mm EXPANSION BOLT	20x20x3 ANGLE BAR	2.4m
325 TO 750	DIA. 6mm	DIA. 6mm EXPANSION BOLT	25x25x3 ANGLE BAR	1.8m
775 TO 1500	DIA. 10mm	DIA. 10mm EXPANSION BOLT	40x40x3 ANGLE BAR	1.8m
1525 TO 2100	DIA. 10mm	DIA. 10mm EXPANSION BOLT	50x50x5 ANGLE BAR	1.8m
2125 OVER	DIA. 10mm	DIA. 10mm EXPANSION BOLT	50x50x6 ANGLE BAR	1.5m



Duct Weight Calculation :

Total duct weight= Duct weight + Insulation weight

Duct weight= Volume x Density

PID Density = 60 kg/m³

Insulation weight= Insulation Volume x Insulation Density

Insulation Density= 32 kg/m³

SCHEDULE FOR DUCT WEIGHT

GI SHEET THICKNESS (MM)	BASIC MASS (KG/MM ²)	UNIT MASS BASE METAL (KG/M ²)	COATING MASS CONSTANT (KG/MM ²)	UNIT MASS OF CUT-LENGTH (KG/M ²)
0.6	7.85	4.71	0.282	4.992
0.8		6.28		6.562
1.0		7.85		8.132
1.2		9.42		9.702

GI DUCT SIZE (mm)	UNIT MASS (KG/M ²)	DUCT WEIGHT (KG/M)	INSULATION WEIGHT (KG/M)	TOTAL WEIGHT (KG/M)
300 x 300	4.992	5.99	2.88	8.87
750 x 750	6.562	19.69	7.20	26.89
1500 x 1500	8.132	48.79	14.40	63.19
2000 x 2000	9.702	77.62	19.20	96.82

SS DUCT SIZE (mm)	UNIT MASS (kg/M ²)	DUCT WEIGHT (KG/M)	INSULATION WEIGHT (KG/M)	TOTAL WEIGHT (KG/M)
300 x 300	7.666	9.19	2.88	12.07
750 x 750		22.99	7.20	30.19
1500 x 1500	9.432	56.59	14.40	63.19
2000 x 2000		75.45	19.20	94.65

ROUND DUCT GAUGE TABLE
(Spiral Seam / Unreinforced)

GI Sheet Thickness	Max. Duct Size (Dia. mm)
	500 Pa Static Pressure
0.5 mm	UP TO 380
0.6 mm	381 – 450
0.8 mm	451 – 650
1.0 mm	651 – 1220
1.2 mm	1221 – 1520

1000 Pa STATIC POS. or NEG.	RECTANGULAR DUCT CONSTRUCTION With Tie Rod Reinforcement			
	1200-mm TDF (Finished length 1100mm)			
	JTR-Joint Tie Rod CTR-Center Tie Rod			
DUCT SIZE \ GI SHEET THICKNESS	1.2mm	1.0mm	0.8mm	0.6mm
UP TO 350				SLIP JOINT
351 TO 450				TDF
451 TO 750			TDF	
751 TO 900		TDF		
901 TO 1050		TDF/CTR		
1051 TO 1350	TDF/CTR			
1351 TO 1800	TDF/CTR/JTR			
1801 & ABOVE	NOT DESIGNED			

CHILLED WATER PIPE :

TABLE-1

A \ B	15-80	100	125	150	200	250	300
15-80	D-10 L-50X50X5 (1 PC)	D-13 L-75X40X5 (1 PC)	D-13 L-75X40X5 (1PC)	D-13 L-75X40X5 (1PC)	D-16 L-75X40X5 (2PCS)	D-16 L-75X40X5 (2PCS)	D-16 L-75X40X5 (2PCS)
100		↓		D-16 L-75X40X5 (2PCS)			
125			↓				
150				↓			
200					↓		
250						↓	
300							↓

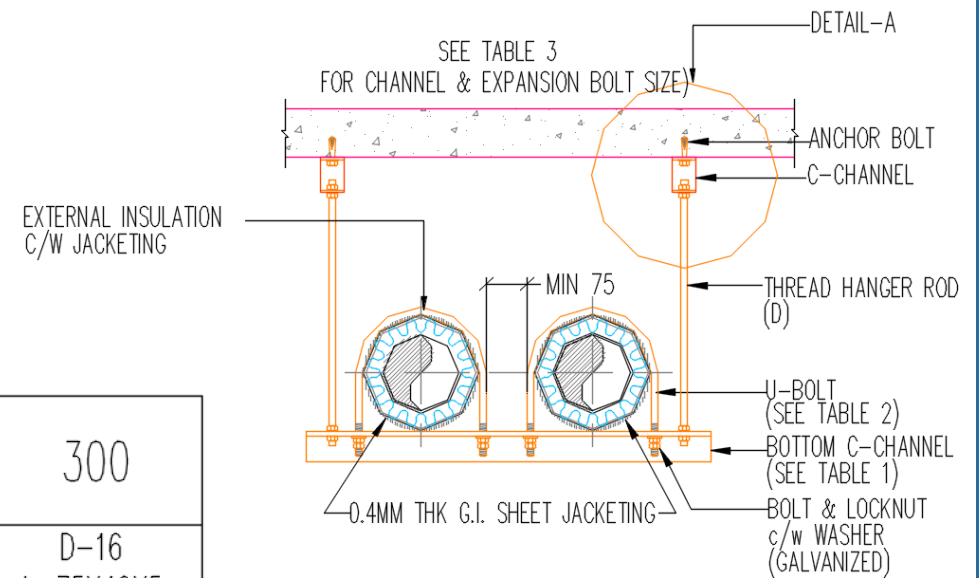
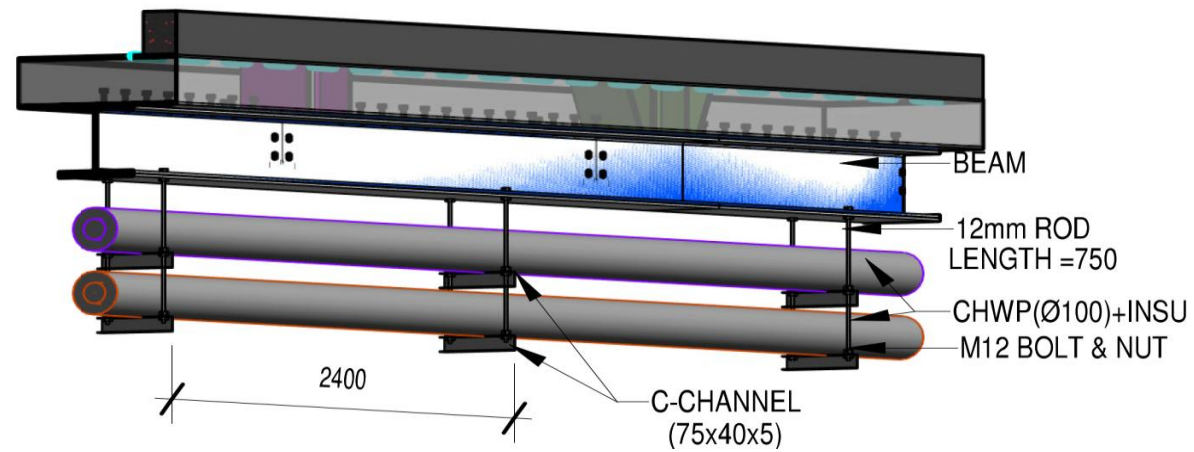


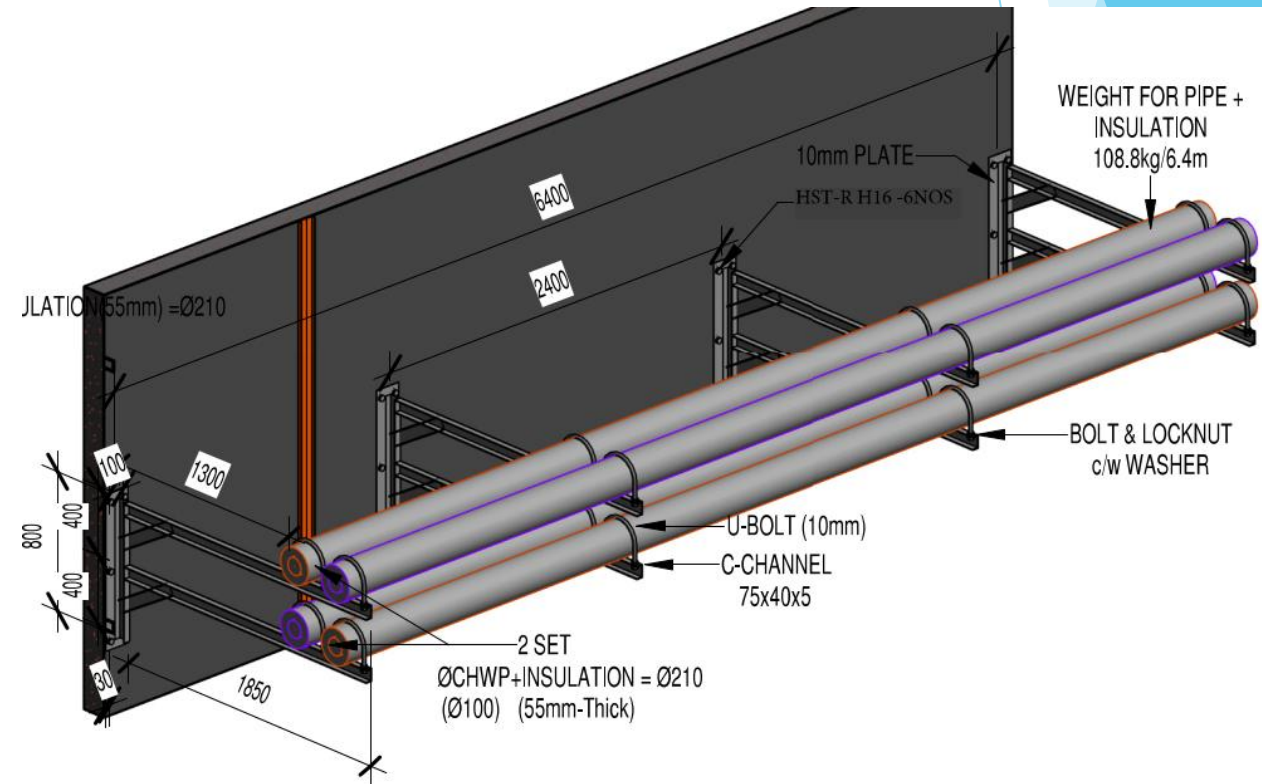
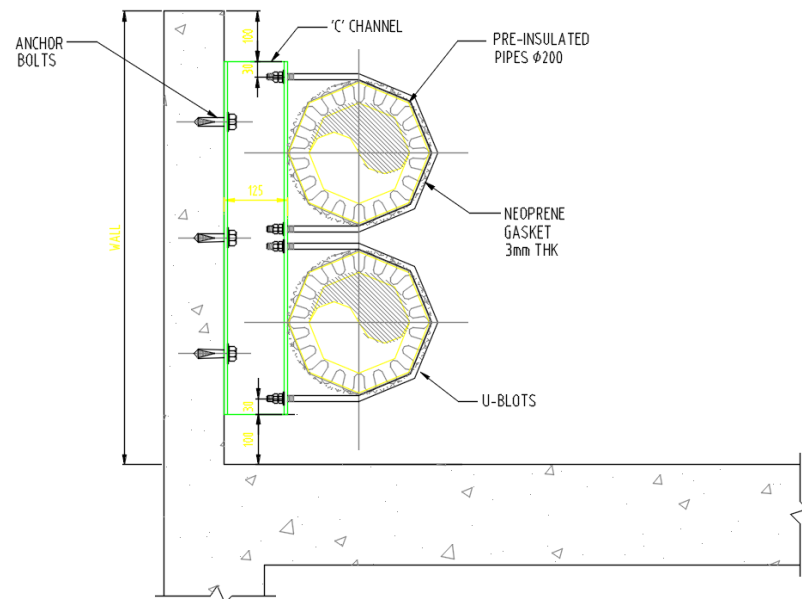
TABLE-2

PIPE SIZE (NORMAL) (MM)	SIZE OF U-BOLT (MM)
15 TO 80	8
100 TO 125	10
150 TO 300	12

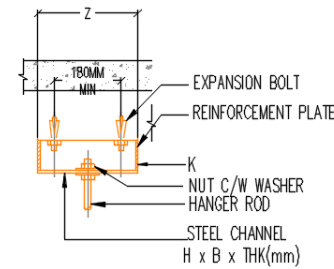
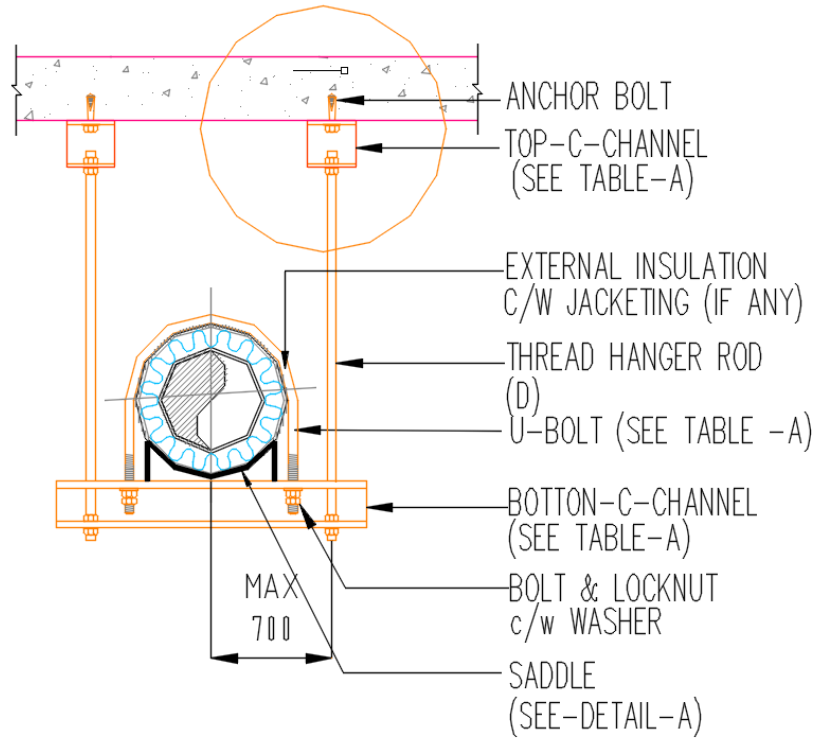
CHILLED WATER PIPE :



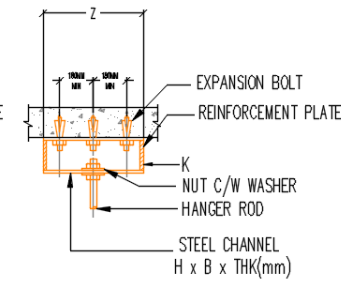
RISER PIPE:



CHILLED WATER PIPE (ABOVE 350mm):



FOR PIPES $\phi 350\text{MM}$ TO $\phi 500$



FOR PIPES $\phi 550$ TOP 750 MM

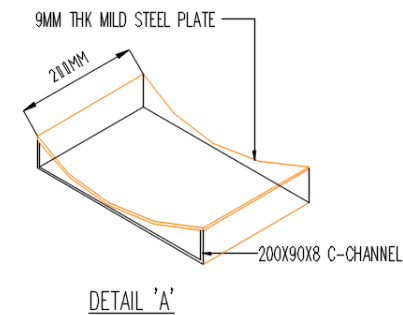


TABLE-A

PIPE SIZE ϕ	TOP CHANNEL		EXPANSION ANCHOR (HILTI HSA)	HANGER ROD D	BOTTOM-CHANNEL	U-BOLT	MAX SPAN
(ϕMM)	K	Z	(MM)	(MM)	(MM)	(MM)	(MM)
$\phi 350-500$	150X75X9TX12.5T	300	M12X110X2	15	150X75X9TX12.5T	12	5.0
$\phi 501-750$	150X75X9TX12.5T	450	M12X110X3	20	150X75X9TX12.5T	12	5.0

Pipe Weight Calculation :



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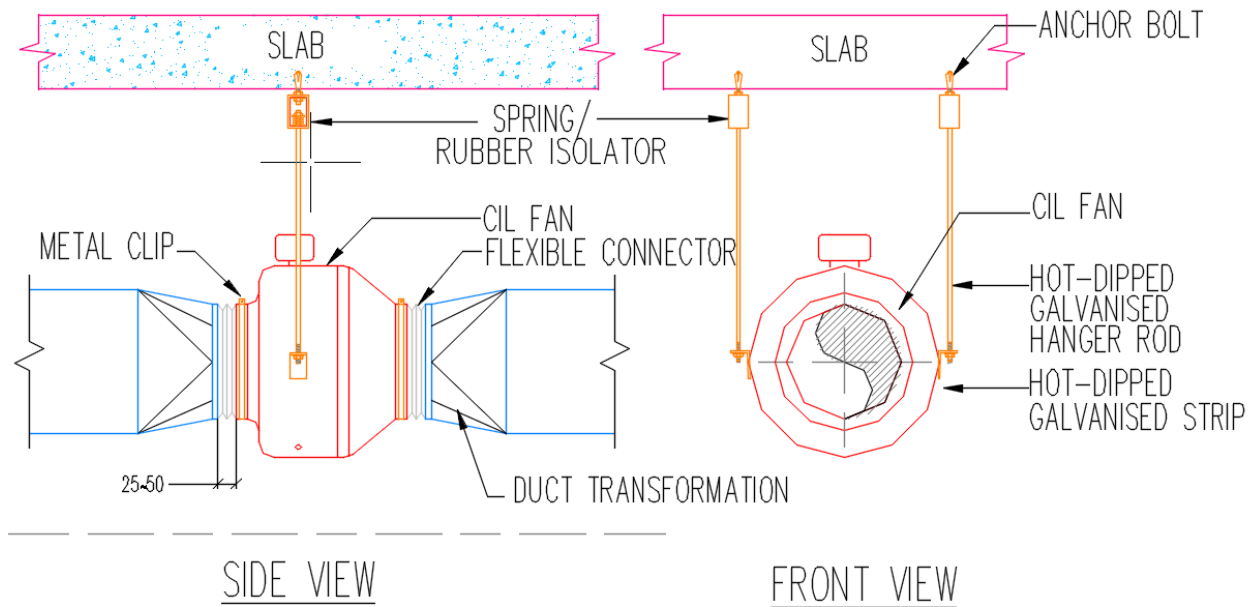


Estimated Pipe Weight

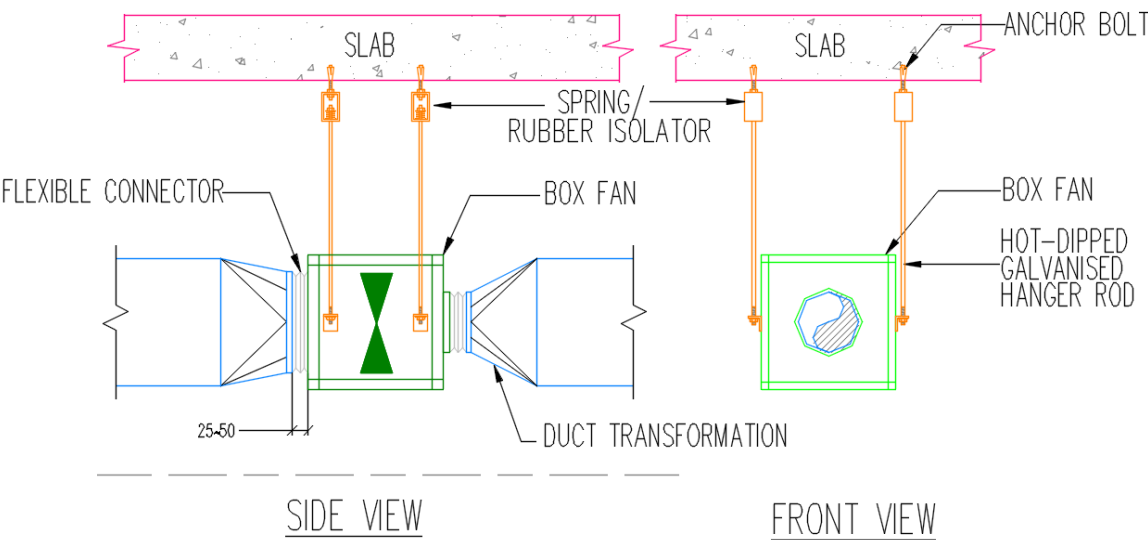
Nominal Pipe Size (mm)	Carrier Pipe Standard	Approximate Weight of Carrier Pipes (kg/6m)	GLENSUL Series No.	Casing Size (mm)	Estimated Weight for Pipe + Water (kg/6m)	Estimated Weight for Pipe + insulation (kg/6m)	Estimated Weight for Pipe + insulation + water (kg/6m)
500	JIS G3452	594.4	65	635	1726.4	701	1843
450	JIS G3452	525.0	65	590	1443.0	634	1552
400	JIS G3452	465.6	65	530	1184.6	561	1280
350	JIS G3452	406.2	65	475	950.2	491	1035
300	JIS G3452	318.0	65	450	756.0	399	837
250	JIS G3452	254.4	65	400	559.4	326	631
200	JIS G3452	180.6	65	350	377.6	243	440
600	API 5L B Sch 40	1532.6	65	735	3090.6	1689	3247
500	API 5L B Sch 40	1100.6	65	635	2176.6	1217	2293
450	API 5L B Sch 40	934.9	65	590	1799.9	1044	1909
400	API 5L B Sch 40	739.9	65	530	1423.9	836	1520
350	API 5L B Sch 40	567.3	65	475	1091.3	652	1176
300	API 5L B Sch 40	478.3	65	450	912.3	558	992
1000	API 5L B Sch XS	1885.4	75	1200	6569.4	2235	6919
950	API 5L B Sch Std	1347.4	65	1100	5564.4	1656	5873
900	API 5L B Sch Std	1275.4	65	1050	5050.4	1570	5345
850	API 5L B Sch Std	1204.9	65	1000	4569.9	1417	4782
800	API 5L B Sch Std	1133.0	65	950	4103.0	1336	4306
750	API 5L B Sch Std	1061.1	65	900	3663.1	1254	3856
700	API 5L B Sch Std	989.2	65	850	3245.2	1173	3429
650	API 5L B Sch Std	917.3	65	800	2853.3	1091	3027
600	API 5L B Sch Std	845.7	65	735	2492.7	1003	2549
500	API 5L B Sch Std	702.9	65	635	1829.9	819	1946
450	API 5L B Sch Std	631.0	65	590	1535.0	740	1644
400	API 5L B Sch Std	559.6	65	530	1267.6	655	1363
350	API 5L B Sch Std	488.0	65	475	1021.0	573	1106
300	API 5L B Sch Std	443.2	65	450	881.2	523	961
250	API 5L B Sch Std	361.7	65	400	666.7	433	738
200	API 5L B Sch Std	255.3	65	350	449.3	317	511
150	API 5L B Sch Std	169.6	50	265	281.6	213	325
125	API 5L B Sch Std	130.6	50	240	207.6	171	248
100	API 5L B Sch Std	96.5	50	215	145.5	126	175
80	API 5L B Sch Std	67.7	50	190	96.7	93	122
65	API 5L B Sch Std	51.8	50	170	70.8	73	92
50	API 5L B Sch Std	32.6	40	140	46.6	50	63
40	API 5L B Sch Std	24.3	40	125	32.3	39	47
32	API 5L B Sch Std	20.3	40	115	26.3	34	40
25	API 5L B Sch Std	15.0	40	115	18.0	29	32
20	API 5L B Sch Std	10.1	40	100	12.1	22	24
15	API 5L B Sch Std	7.6	40	100	8.6	20	21
150	BS1387 Class B	118.2	50	265	232.2	162	276
125	BS1387 Class B	99.6	50	240	178.6	140	219
100	BS1387 Class B	73.2	50	215	125.2	102	154
80	BS1387 Class B	50.2	50	190	81.2	75	106
65	BS1387 Class B	38.6	50	170	60.6	60	82
50	BS1387 Class B	30.2	40	140	43.2	47	60
40	BS1387 Class B	21.4	40	125	30.4	36	45
32	BS1387 Class B	18.6	40	115	24.6	32	38
25	BS1387 Class B	14.5	40	115	18.5	28	32
20	BS1387 Class B	9.4	40	100	12.4	21	24
15	BS1387 Class B	7.3	40	100	8.3	19	20
150	BS1387 Class C	127.8	50	265	239.8	172	284
125	BS1387 Class C	107.4	50	240	185.4	148	226
100	BS1387 Class C	87.0	50	215	138.0	116	167
80	BS1387 Class C	61.8	50	190	90.8	87	116
65	BS1387 Class C	47.6	50	170	68.6	69	90
50	BS1387 Class C	37.1	40	140	50.1	54	67
40	BS1387 Class C	28.2	40	125	34.2	41	49
32	BS1387 Class C	22.8	40	115	27.8	37	42
25	BS1387 Class C	17.6	40	115	20.6	32	35
20	BS1387 Class C	11.2	40	100	13.2	23	26
15	BS1387 Class C	8.6	40	100	9.6	21	22

Equipment Supporting Detail:

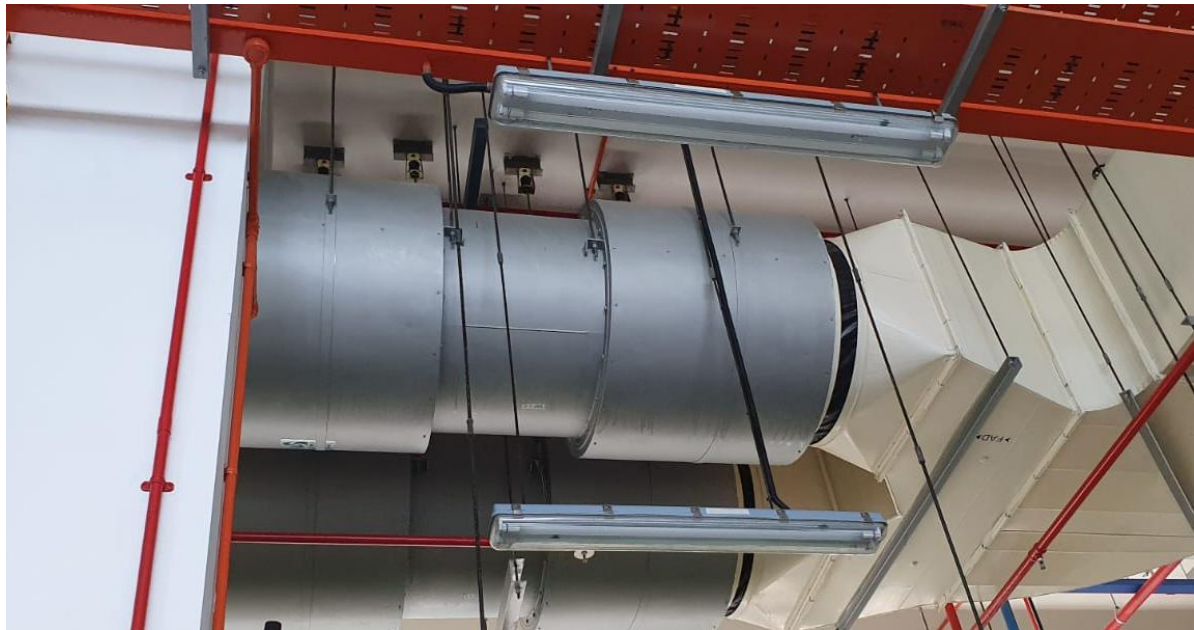
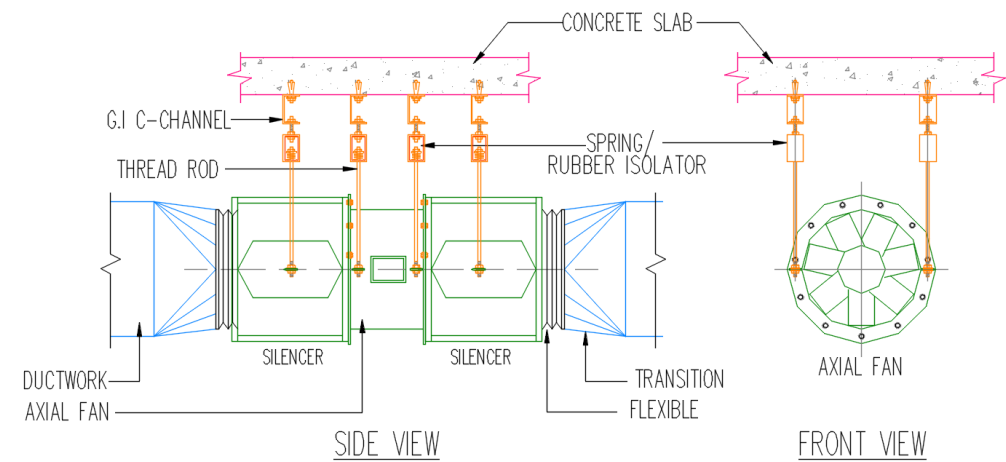
MV FAN(CIL):



MV FAN(BOX):



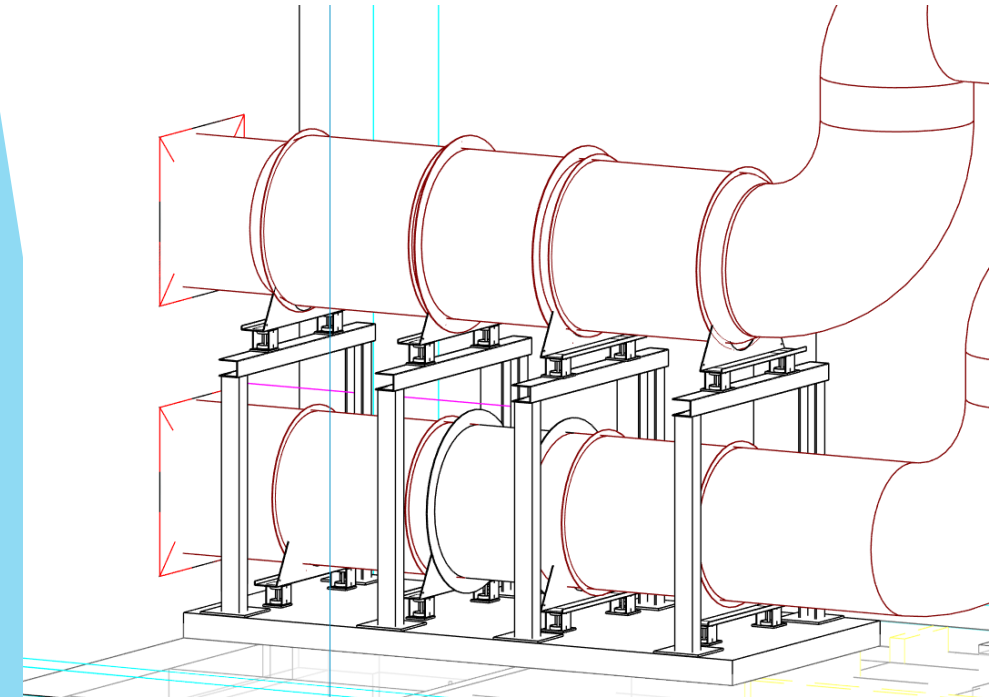
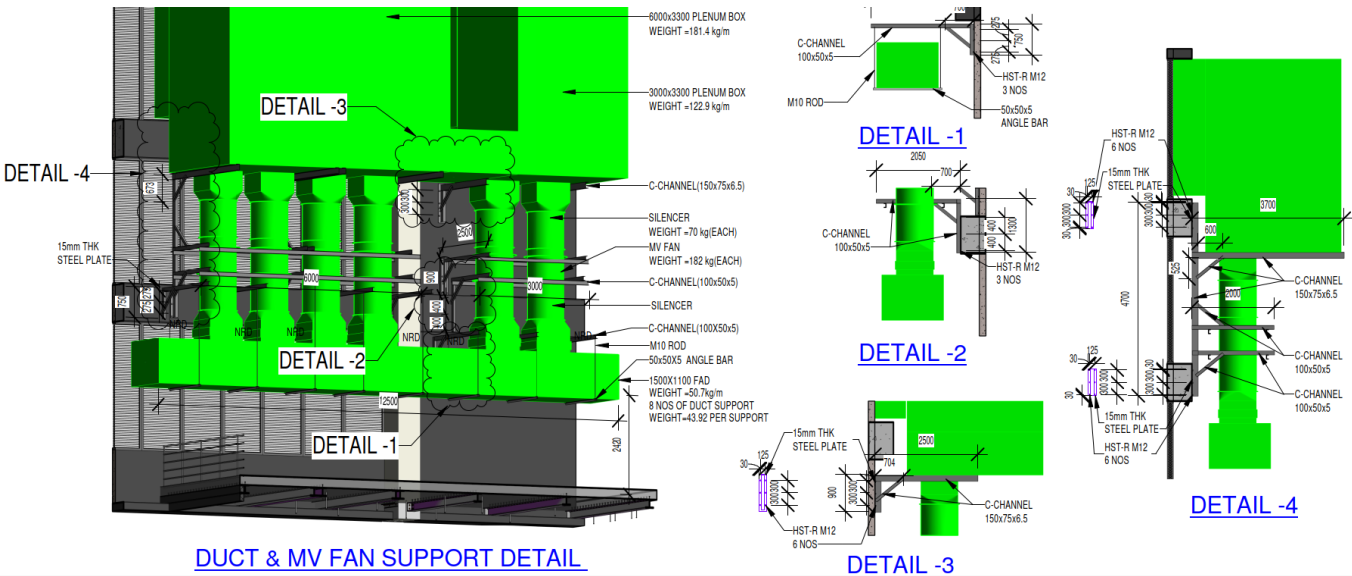
MV FAN(AXIAL):



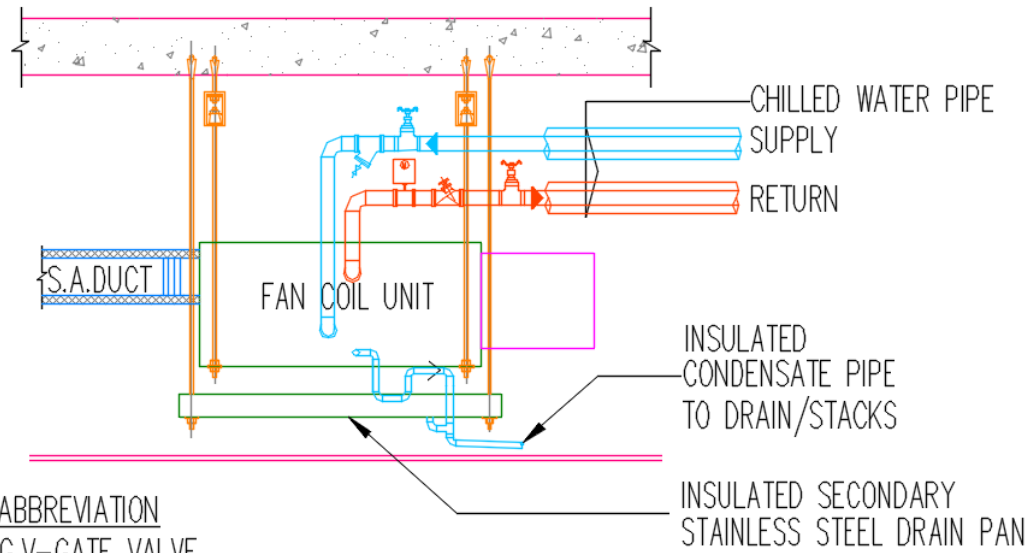
HANGER SIZING

FAN DIMENSION ϕ mm(inch)	FANS HANGER AND SUPPORT DIMENSION		
	HANGER ROD (ϕ mm)	C-CHANNEL (mm)	ANCHOR BOLT (mm x Nos.)
305 (12)	10	100 x 50 x 5	M10 x 4
381 (15)	10	100 x 50 x 5	M10 x 4
481 (19)	10	100 x 50 x 5	M10 x 4
533 (21)	12	100 x 50 x 5	M12 x 4
610 (24)	12	100 x 50 x 5	M12 x 4
762 (30)	16	100 x 50 x 5	M10 x 2 x 4
965 (38)	16	125 x 65 x 6	M12 x 2 x 4
1219 (48)	18	125 x 65 x 6	M16 x 2 x 4
1524 (60)	18	150 x 75 x 8	M16 x 2 x 4
1905 (75)	18	150 x 75 x 8	M16 x 2 x 4

MV FAN(AXIAL):



FCU(Ducted):



ABBREVIATION

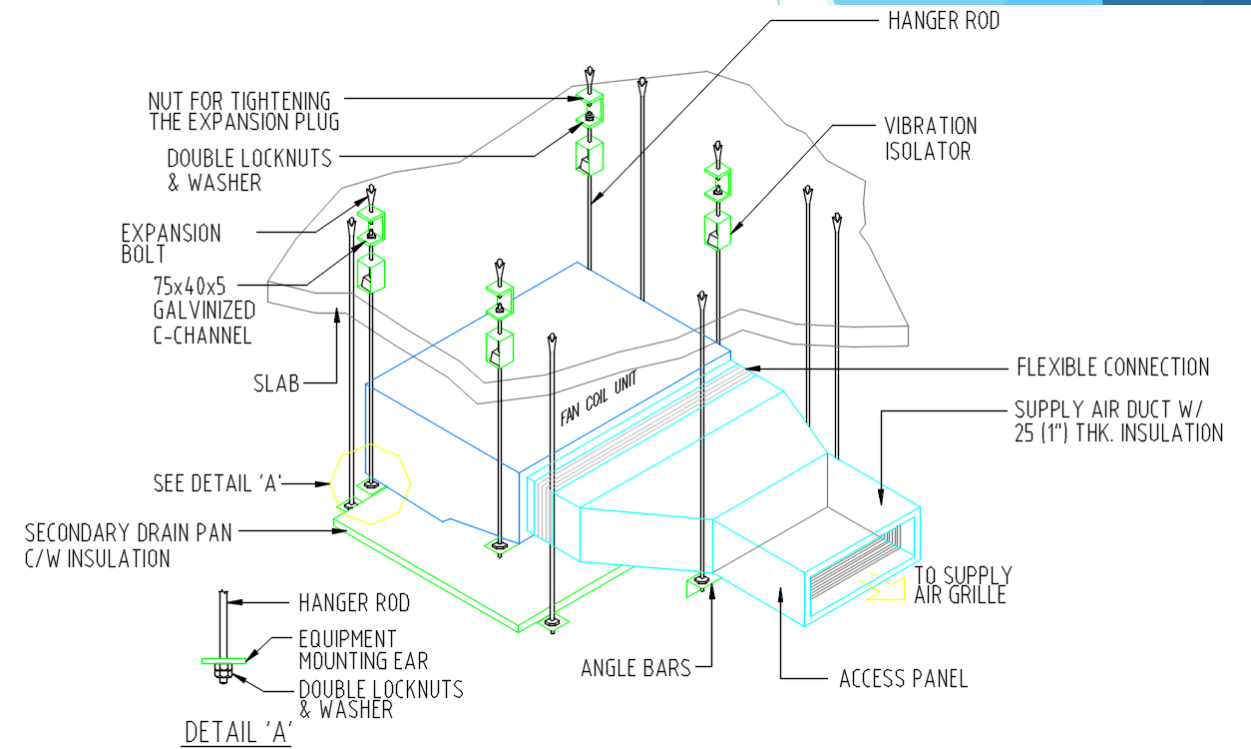
G.V-GATE VALVE

M.V-MODULATING VALVE

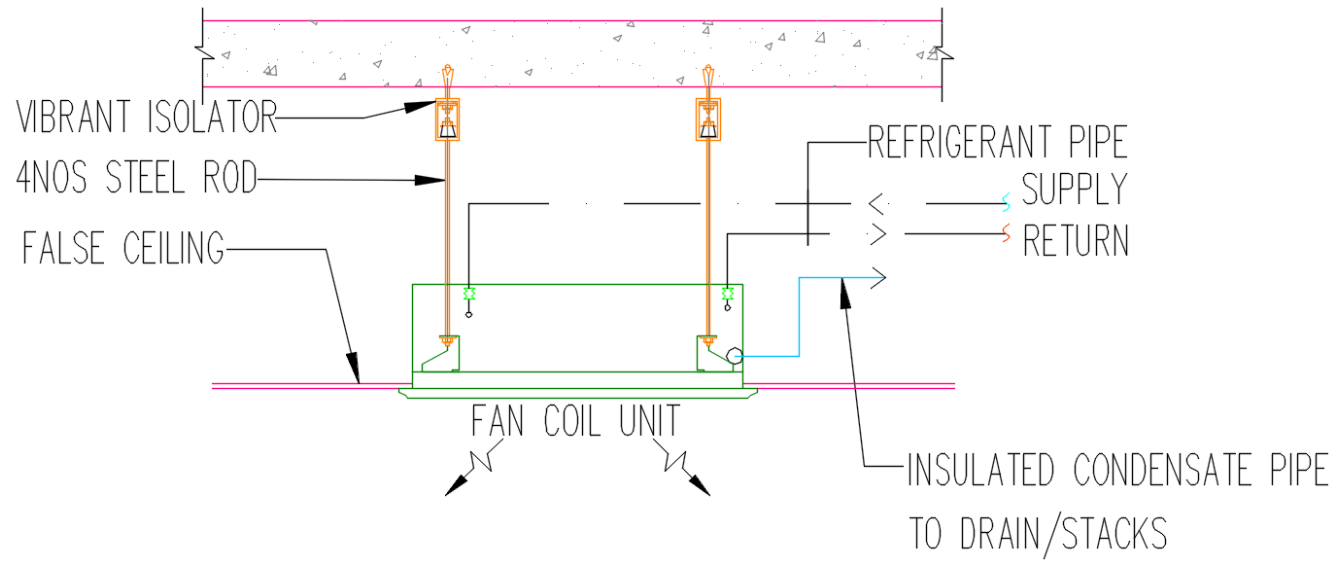
F.C-FLEXIBLE CONNECTOR

S-Y-STRAINER

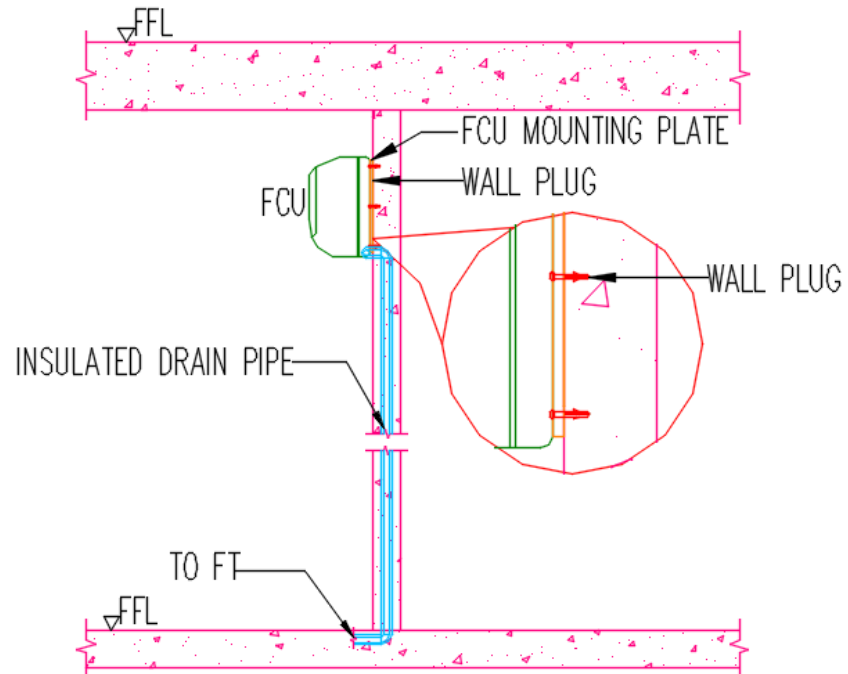
B.V-BALANCING VALVE



FCU(Cassette):

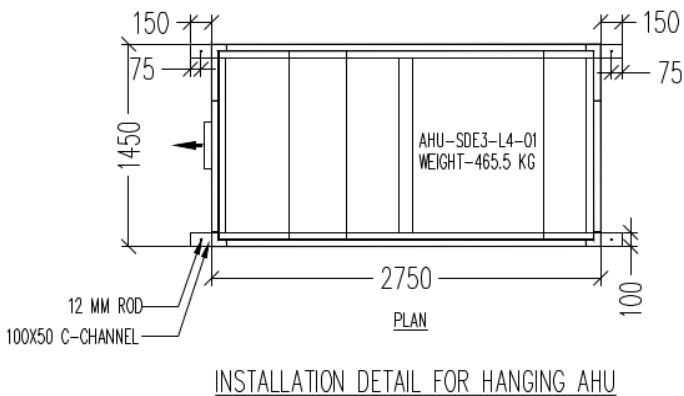
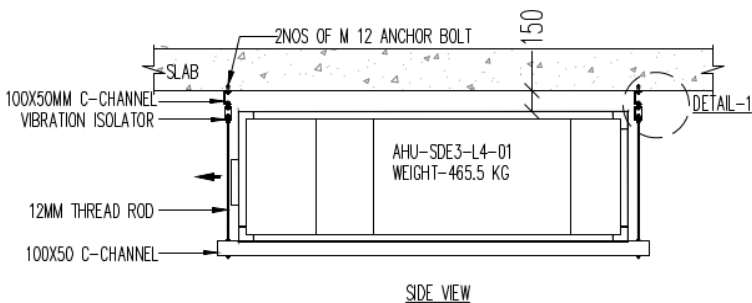


FCU(Wall mounted):



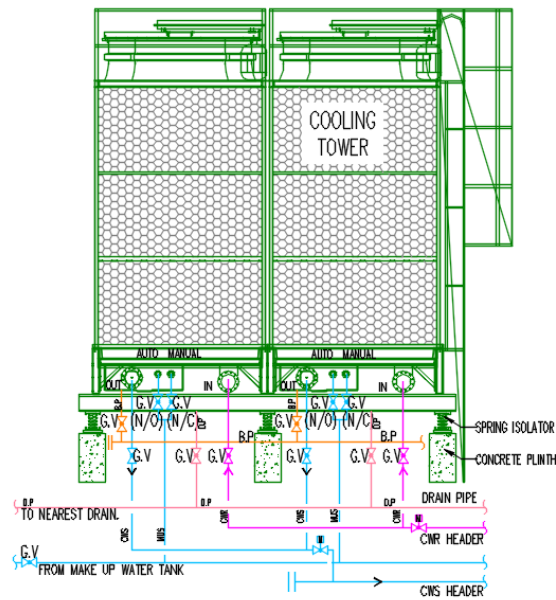
Equipment Installion

AHU:

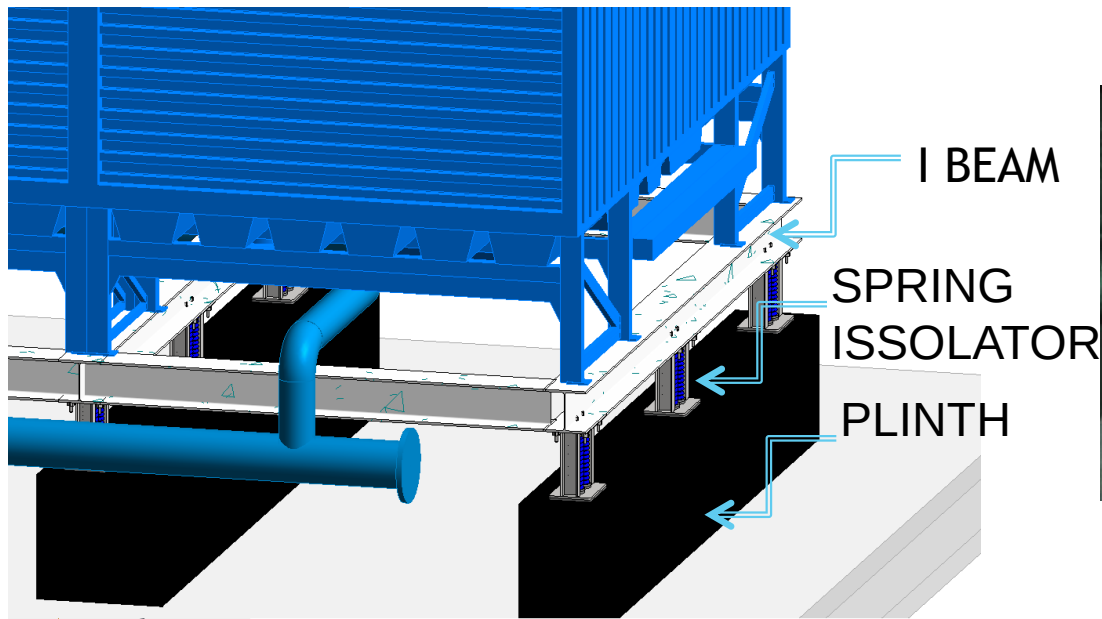


Equipment Installion

COOLING TOWER:

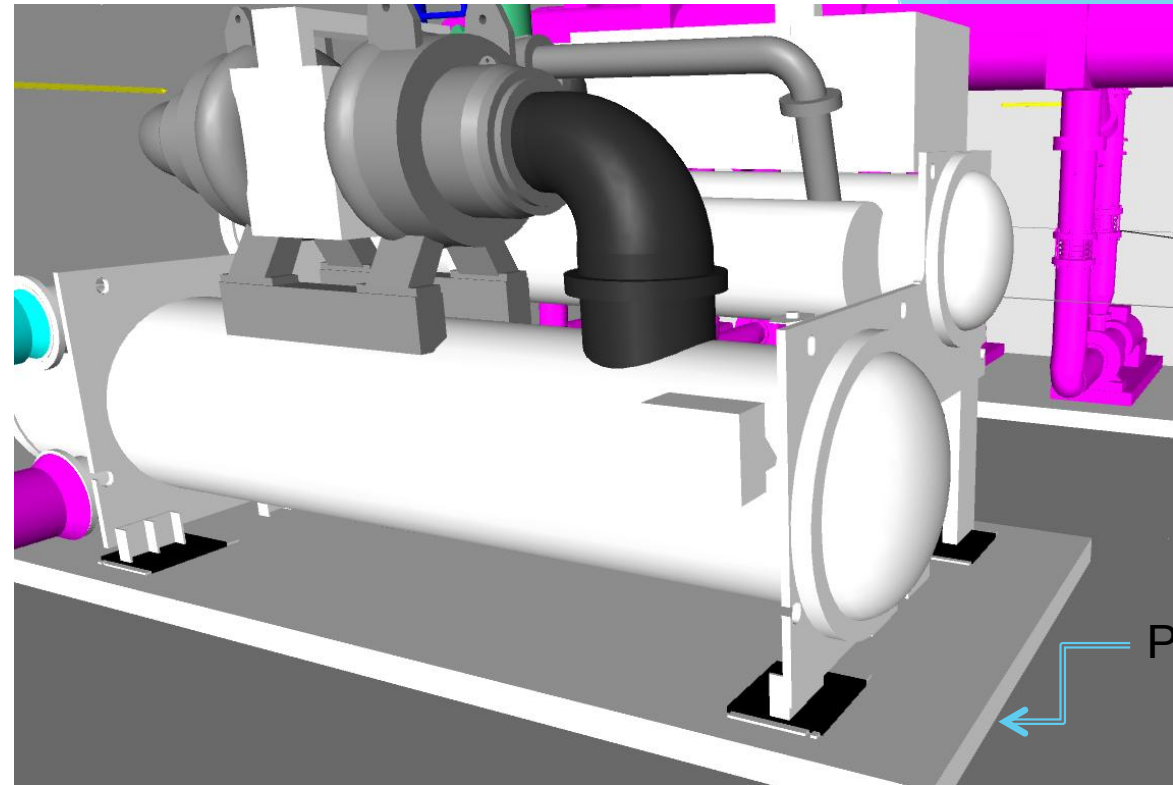
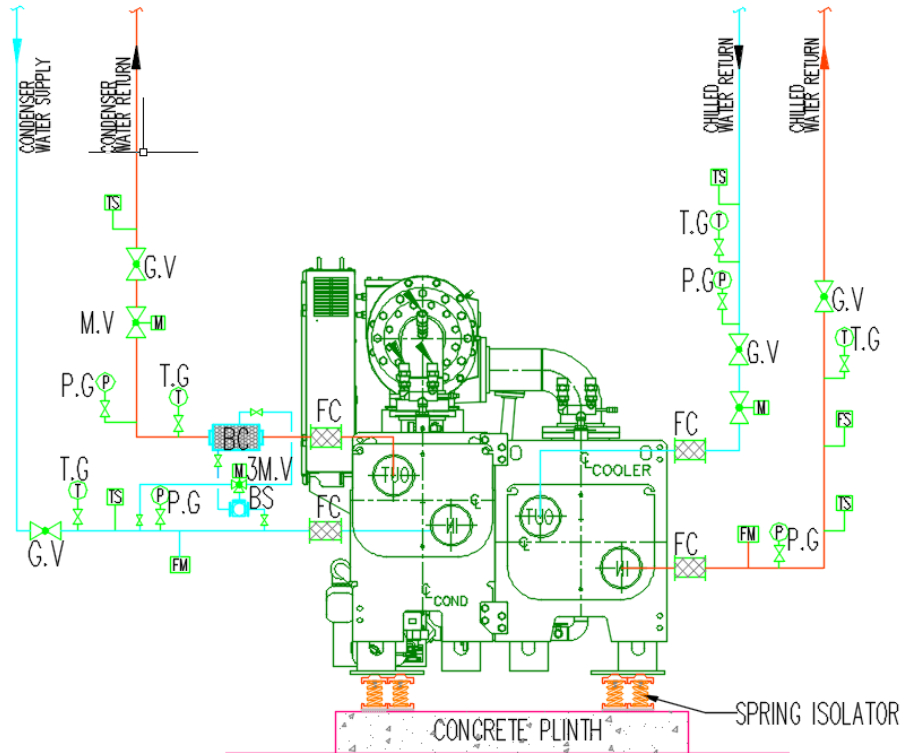


ABBREVIATION
G.V.-GATE VALVE
CWS-CONDENSER WATER SUPPLY
CWR-CONDENSER WATER RETURN
MUS-MAKE UP WATER SUPPLY
B.P.-BALANCING PIPE
D.P.-DRAIN PIPE



Equipment Installion

CHILLER:



INERTIA BLOCK :

